

## HOW POPULATIONS EVOLVE

# Module map

- Module 13: How Populations Evolve
- Connected lab: lab-14\_how-populations-evolve.md
- Use this deck as the visual path through the module's evidence and retrieval work.

## Module 13

How Populations Evolve

1

Allele frequencies as the unit of change

2

Sources of variation

3

Five mechanisms of microevolution

4

Genetic drift and selection

5

Hardy-Weinberg as a baseline

### Lab evidence

lab-14\_how-populations-evolve.md

## HOW POPULATIONS EVOLVE

# Learning objectives

- Define microevolution using allele frequencies.
- Compare the five mechanisms that change allele frequencies.
- Distinguish bottleneck and founder effects.
- Interpret directional, stabilizing, and disruptive selection.
- Use Hardy-Weinberg as a non-evolving baseline.

**OBJECTIVE 1**

Define microevolution using allele frequencies.

**OBJECTIVE 2**

Compare the five mechanisms that change allele frequencies.

**OBJECTIVE 3**

Distinguish bottleneck and founder effects.

**OBJECTIVE 4**

Interpret directional, stabilizing, and disruptive selection.

**OBJECTIVE 5**

Use Hardy-Weinberg as a non-evolving baseline.

## HOW POPULATIONS EVOLVE

# Topic sequence

- Allele frequencies as the unit of change: Microevolution is change in allele frequencies across generations.
- Sources of variation: Mutation creates new variation, while recombination reshuffles existing variation.
- Five mechanisms of microevolution: Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.
- Genetic drift and selection: Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.
- Hardy-Weinberg as a baseline: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

01

## Allele frequencies as the unit of change

Microevolution is change in allele frequencies across generations.

02

## Sources of variation

Mutation creates new variation, while recombination reshuffles existing variation.

03

## Five mechanisms of microevolution

Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.

04

## Genetic drift and selection

Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.

05

## Hardy-Weinberg as a baseline

Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

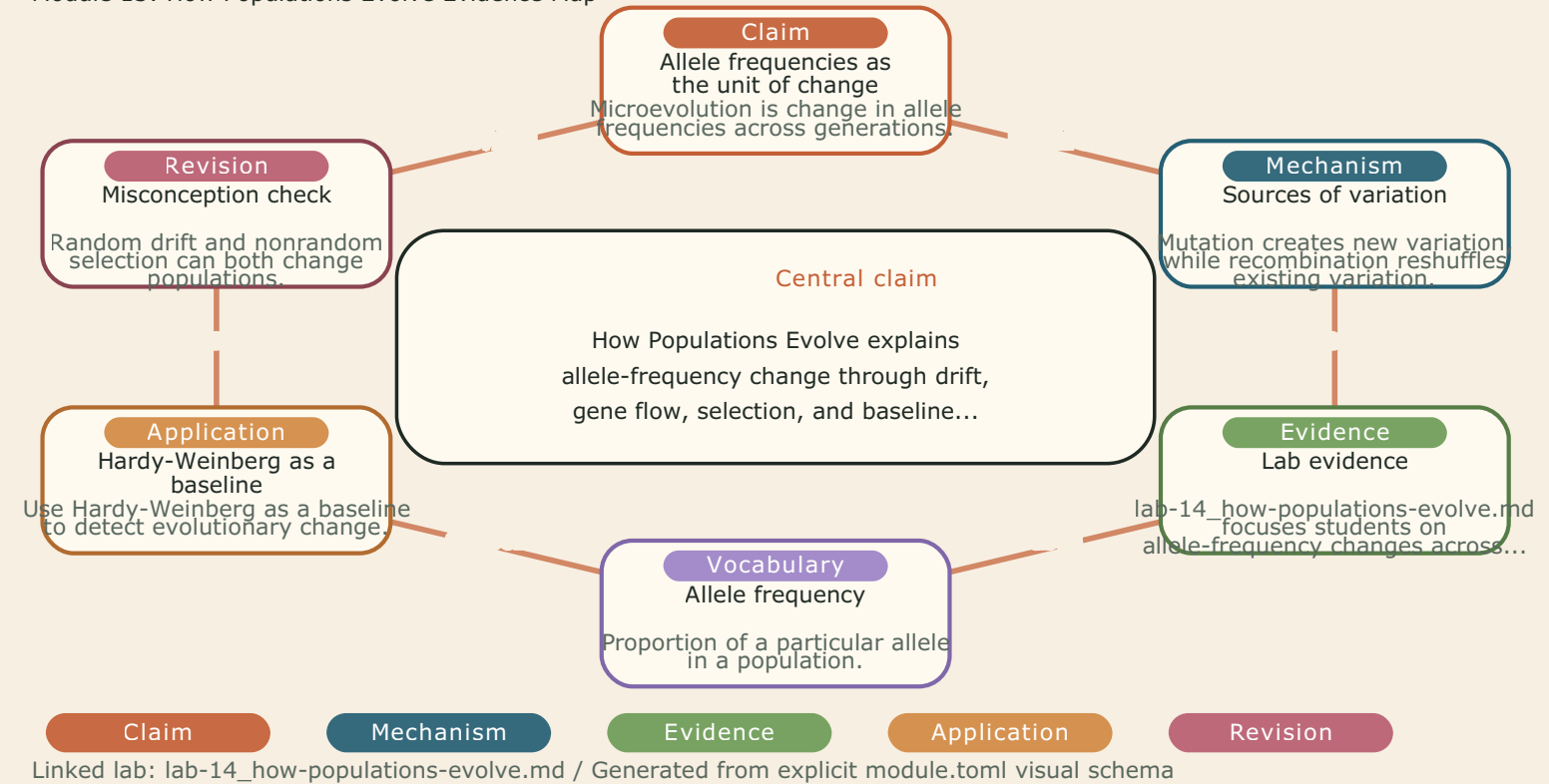
## HOW POPULATIONS EVOLVE

# Module 13: How Populations Evolve Evidence Map

- Module focus: How Populations Evolve
- Central claim: How Populations Evolve explains allele-frequency change through drift, gene flow, selection, and baseline models.
- Key nodes to connect: Allele frequencies as the unit of change, Sources of variation, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-14\_how-populations-evolve.md supplies an evidence surface for the map.

## Module 13 / Concept Map

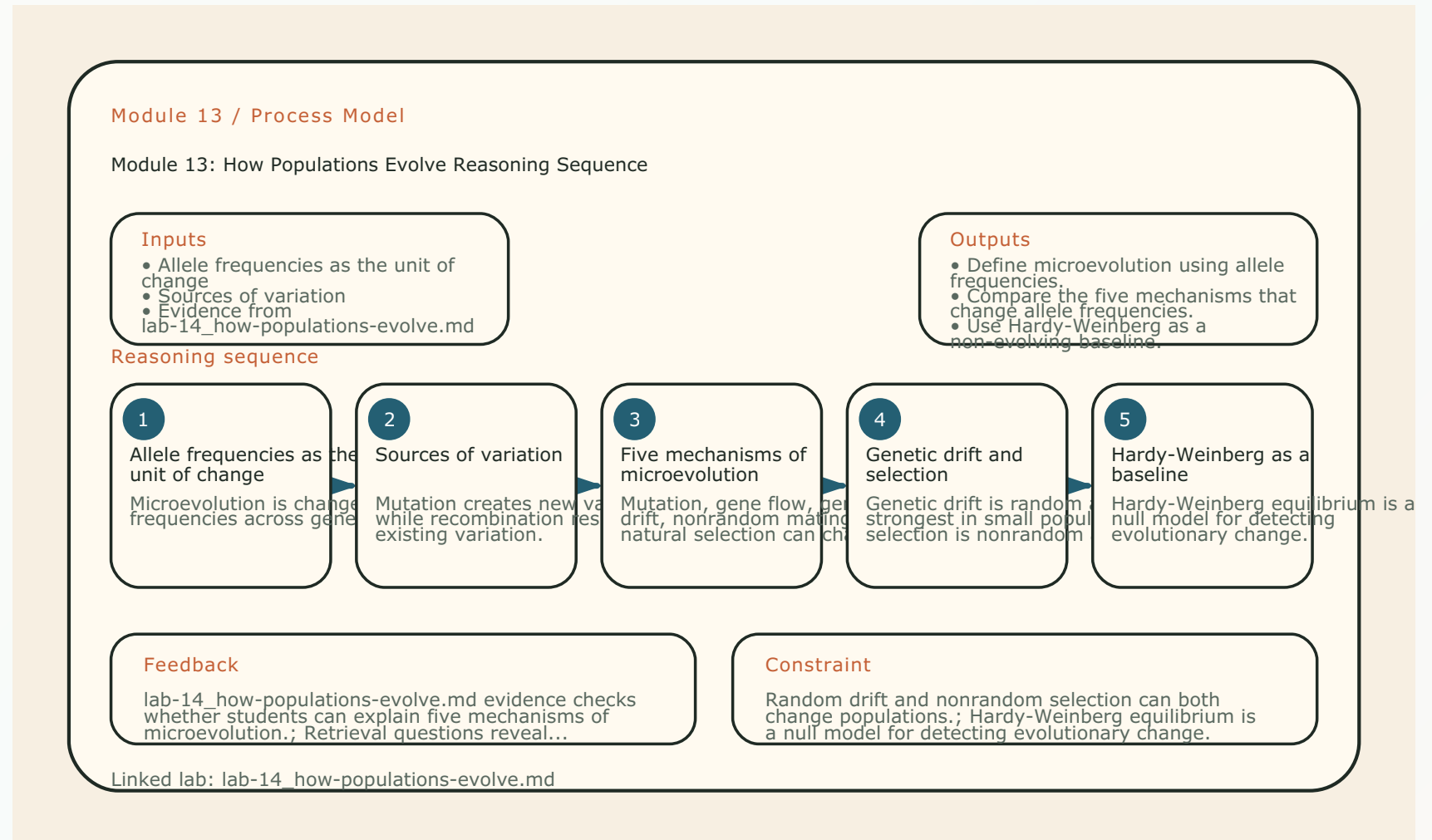
## Module 13: How Populations Evolve Evidence Map



## HOW POPULATIONS EVOLVE

# Module 13: How Populations Evolve Reasoning Sequence

- Module focus: How Populations Evolve
- Trace the model: Allele frequencies as the unit of change -> Sources of variation -> Five mechanisms of microevolution -> Genetic drift and selection
- Inputs become outputs: Allele frequencies as the unit of change, Sources of variation -> Define microevolution using allele frequencies., Compare the five mechanisms that change allele frequencies.
- Feedback check: lab-14\_how-populations-evolve.md evidence checks whether students can explain five mechanisms of microevolution.
- Lab connection: lab-14\_how-populations-evolve.md gives students a process to observe or test.



## HOW POPULATIONS EVOLVE

# Terms as evidence handles

- Allele frequency: Proportion of a particular allele in a population.
- Microevolution: Allele-frequency change across generations.
- Mutation: DNA sequence change that can create new alleles.
- Gene flow: Movement of alleles between populations.
- Genetic drift: Random allele-frequency change, strongest in small populations.
- Bottleneck effect: Drift after a sharp population reduction.

**ALLELE FREQUENCY**

Proportion of a particular allele in a population.

**MICROEVOLUTION**

Allele-frequency change across generations.

**MUTATION**

DNA sequence change that can create new alleles.

**GENE FLOW**

Movement of alleles between populations.

**GENETIC DRIFT**

Random allele-frequency change, strongest in small populations.

**BOTTLENECK EFFECT**

Drift after a sharp population reduction.

## HOW POPULATIONS EVOLVE

# Lab connection

- Lab file: lab-14\_how-populations-evolve.md
- Lab evidence should test or illustrate: Five mechanisms of microevolution
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

**QUESTION**

Allele frequencies as the unit of change

**EVIDENCE**

lab-14\_how-populations-evolve.md

**CLAIM**

Define microevolution using allele frequencies.

## HOW POPULATIONS EVOLVE

# Contrast check

- Do not stop at: Microevolution is change in allele frequencies across generations.
- Stronger explanation: Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.
- The goal is to move from a named idea to a testable biological explanation.

**SURFACE**

Microevolution is change in allele frequencies across generations.

**DEEPER**

Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

## HOW POPULATIONS EVOLVE

# Module 13: How Populations Evolve Retrieval and Lab Check

- Module focus: How Populations Evolve
- Retrieval prompt: What is an allele frequency?
- Answer check: Define microevolution using allele frequencies.
- Required terms: Allele frequency, Microevolution, Mutation
- Lab connection: lab-14\_how-populations-evolve.md; lab-14\_how-populations-evolve.md supplies evidence for allele-frequency changes across generations.

## Module 13 / Retrieval Card

Module 13: How Populations Evolve

Retrieval and Lab Check

## Routine

Cover notes -> answer aloud  
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

Q1

What is an allele frequency?

Check: Define microevolution using allele frequencies.

Q2

How is microevolution measured?

Check: Compare the five mechanisms that change allele frequencies.

Q3

Which mechanisms can change allele frequencies?

Check: Distinguish bottleneck and founder effects.

Q4

How does mutation differ from gene flow?

Check: Interpret directional, stabilizing, and disruptive selection.

## Terms to use

Allele frequency, Microevolution, Mutation, Gene flow, Genetic drift

## Lab connection

lab-14\_how-populations-evolve.md supplies evidence for allele-frequency changes across generations.

Intro biology routine: claim, evidence, reasoning, revision.

## HOW POPULATIONS EVOLVE

# Practice quiz bridge

- 1. Microevolution is defined as:
- 2. Genetic drift has its strongest effect in:
- 3. A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:
- 4. The Hardy-Weinberg principle is useful mainly because it:

**QUESTION 1****Answer first**

Microevolution is defined as:

**QUESTION 2****Answer first**

Genetic drift has its strongest effect in:

**QUESTION 3****Answer first**

A few birds colonize a new island and start a population with limited variation. This reduced variation is due to:

**QUESTION 4****Answer first**

The Hardy-Weinberg principle is useful mainly because it:

## HOW POPULATIONS EVOLVE

# Synthesis and exit ticket

- Exit claim: explain allele frequencies as the unit of change using evidence from lab-14\_how-populations-evolve.md.
- Must include: Allele frequency, Microevolution, Mutation.
- Revision prompt: what evidence would change your explanation?

**CLAIM**

Allele frequencies as the unit of change

**EVIDENCE**

lab-14\_how-populations-evolve.md

**REVISION**

What would change your mind?